

Abstract

Carcinus maenas is a globally invasive species that outcompetes native species such as *Hemigrapsus* spp. Both species are adapted to withstand wide ranges of environmental factors; however, salinities within water bodies such as Puget Sound are projected to undergo a long-term, climate change-forced freshening. Physiological responses induced by salinity stress expressed in crab species is poorly characterized over temporal scales. Further, little research has compared the stress-induced physiological responses between *C. maenas* and *Hemigrapsus* spp. This study subjected populations of these species to three-week sustained salinity treatments of 15, 35, and 45PSU and measured common physiological traits such as righting time, respiration, and hemolymph osmolality over time. Both species experienced heightened stress in 15PSU, represented as decreased righting times, adding to knowledge of how fresher waters impact stress responses and evasive actions in crabs. *C. maenas* physiologically tolerated salinity extremes more effectively than *Hemigrapsus* spp. This finding was expected, as extensive research has shown enhanced environmental tolerance in *C. maenas* for invasion into non-native regions. These results show that a freshening of Puget Sound may cause enhanced stress responses in these crab species, with potential for *C. maenas* becoming a larger invasive threat to *Hemigrapsus* spp. This study provides initial findings of how native and invasive species respond to changing environmental conditions. It is imperative to continue characterizing physiological responses to climate-forced environmental factors to further understand how populations may change in the future and how conservation efforts can be specifically tuned to benefit crab species based on research findings.

Introduction

Invasive species such as *Carcinus maenas* (European green crab; EGC) can cause many adverse effects on native species such as *Hemigrapsus oregonensis* and *Hemigrapsus nudus* (hairy shore crab; HSC) including habitat loss and displacement, food scarcity, and evolutionary shifts (Rivera et al., 2011). Previous studies have shown that EGC populations significantly decreased the abundance of HSC populations (Rivera et al., 2011). EGC originate from northwestern Europe and the United Kingdom and have been distributed globally (Hill et al., 1991). These crabs have adapted to inhabit a wide range of habitats, including northwestern Canada, southern Australia, and many places in between (Howard et al., 2018). This wide breadth of inhabitants is evidence of the wide range of environmental conditions these crabs can survive. For example, research has found that EGCs have a preferred thermal tolerance between 6-23°C with poorly characterized thermal limits (Kelley et al., 2011). Contrastingly, HSCs are much more restricted to a specific geographic region, decreasing the need for wide environmental tolerance. This difference in tolerance and potential benefit to EGC physiology has been well characterized for temperature ranges; however, other environmental conditions such as changes in salinity are not as well studied, yet are still relevant in understanding how these animals and their interactions change through shifting conditions due to processes such as climate change (Dehnel, 1960; Jimenez et al., 2025; Kelley et al., 2011; Rivera et al., 2011; Symons, 1964; Todd & Dehnel, 1960).

The Puget Sound, located in western Washington state, is an estuarine system with several large river deltas. These rivers feed the Sound's shorelines and surface waters with freshwater seasonally melted from glaciers and snowpack in the Cascade Mountains. Seasonal cycles in freshwater input have been a long-standing process for the Puget Sound and its ecosystems;

however, climate change is threatening a slow freshening of the water body (Walker et al., 2022). This freshening is predicted to be caused by increased glacier and snowpack melt rates, creating a sustained change of ~1PSU in some areas throughout Puget Sound, measured between the short term (2020-2050) and the long term (2070-2100) (Walker et al., 2022). While the crabs examined in this study can withstand large swings in salinity for hours at a time due to their intertidal habitats, creating a more permanent, observable change in salinity may have an impact on physiology and interaction behaviors in ways not yet understood.

It is important to characterize how a freshening of Puget Sound may impact the physiological responses and survivability of HSCs and EGCs to have an idea of how these species may respond to these long-term environmental changes, both individually and as an interaction with each other. For example, if EGCs can better adapt to fresher waters, it is important to predict how much the invasive species will outcompete native species to anticipate further changes to ecosystem dynamics, conservation efforts, and potential fishery implications of other, more commercially valuable native species. To characterize this, populations of EGC and HSC were subjected to various salinity treatments held at a constant for three weeks. Each week, physiological traits (righting time, respiration, and hemolymph osmolality) were compared between species, salinity treatments and weeks. With each of these physiological traits, it was hypothesized that a lower salinity would result in an increased righting time, respiration rate, and a decreased hemolymph osmolality. Lower salinities may pique stress in crabs since they are more likely to frequently experience high saline environments in the wild. Further, it was hypothesized that HSCs would be able to right themselves faster than EGCs due to their small anatomy and seemingly finer motor skills. As a final hypothesis, it was predicted that when salinity is higher than typical environmental levels, HSCs would have a higher osmolality than EGCs and vice versa. All hypotheses support the idea that HSCs are less suited for coping with salinity changes and are more vulnerable to being outcompeted by EGCs in the future.

Methods

For the purposes of this study, *H. oregonensis* and *H. nudus* were treated as physiologically indistinguishable, both contributing to the HSC population. 15, 35, and 45PSU salinity treatments were set up to simulate environmental extremes (15 and 45PSU) and ambient seawater salinities (35PSU). Four HSCs of similar masses were chosen per salinity treatment. Each group of four was placed in a small glass jar with an air stone strung through the small hole in the jar's lid. The jar was placed in a larger tub that housed one EGC per salinity treatment. Each of the three tubs were placed in a water table with a sustained ambient temperature as a control variable.

Mass and righting time

Each of the three weeks of the study, physiological observations started with massing each crab and testing its righting time. Each crab was placed on a standard scale to measure its mass. Each crab was then placed and held on its dorsal carapace, released, and allowed to right itself against a monitored stopwatch. Righting time was adjusted by each crab's respective mass.

Respiration

Resazurin was used as a proxy for measuring respiration. Resazurin is redox-sensitive, blue dye which is reduced to resorufin (pink in color and highly fluorescent) at a rate that matches the

cellular respiration of the animal in the dye. 6-well plates were used to contain HSCs, while small glass jars were used to contain EGCs. Enough resazurin was added to each well/jar to submerge the crab and ensure that it was breathing the dye solution. One control per containment method was added. Aliquots of 200 μ L were taken from each sample at 0, 30, and 60 minutes. Resorufin fluorescence was then measured using a spectrophotometer plate. Each crab underwent two rinses to wash away any residual resazurin before being returned to their respective salinity treatments.

Hemolymph extractions and osmolality

Osmolality can be used as a proxy for understanding stress and how well the animal is able to regulate its own osmolality against external salinity conditions. At the start of the experiment, two non-experimental HSCs had a hemolymph extraction performed to provide a baseline osmolality that could be compared to all experimental crabs' osmolalities taken at the end of the experiment. Each extraction was performed with a hypodermic needle inserted into the leg joint of the crab. The needle was used to collect a small amount of hemolymph which was processed using an osmometer.

Experimental notes

Due to incorrect experimental setup in week one, all crabs in salinity treatments 35 and 15PSU died due to anoxia. These crabs were replaced for the remainder of the study. One HSC in the 35PSU treatment molted prior to week two measurements and did not survive the process (not replaced). One HSC in the 15PSU treatment died prior to week three measurements (not replaced). The EGC in the 35PSU treatment molted prior to week 3 measurements and survived.

Results

In EGCs, the 35PSU treatment saw a righting time increase over time while the other salinity treatments decreased with time. In HSCs, 15PSU resulted in a decrease in righting time while 35 and 45PSU resulted in a peak in righting time in weeks 2 and 3, respectively. 15PSU resulted in the overall lowest righting time across both species. HSCs presented higher average mass-adjusted righting times across each week and salinity treatment compared with EGCs (Fig. 1).

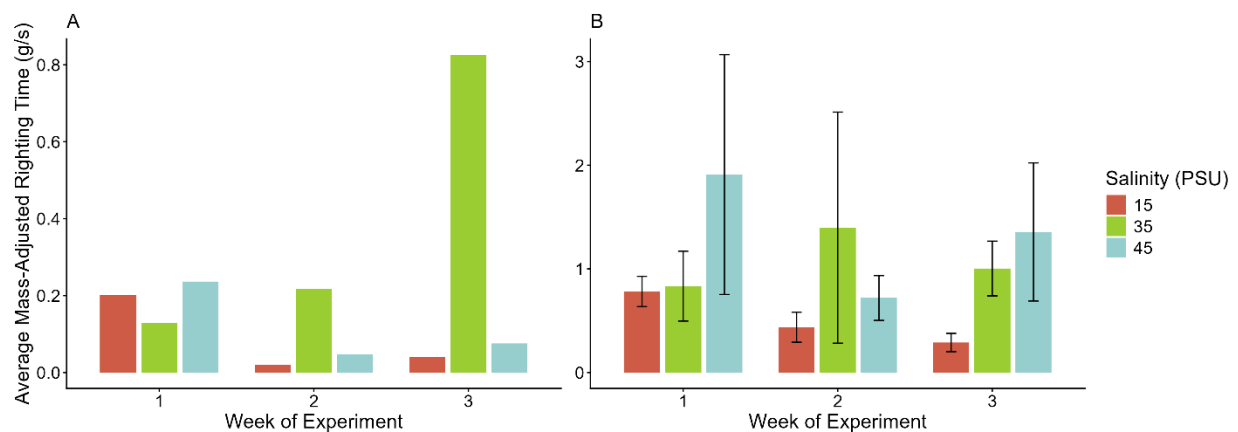


Figure 1. Average mass-adjusted righting time of EGCs (A) and HSCs (B) in salinity treatments of 15, 35, and 45PSU over a 3-week experimental period. One EGC was present in each salinity condition, making error bars inappropriate. Standard error was used to create error bars for the HSC data. Note the change in scale on the y-axis between panels.

Little variability was observed in EGC respiration over time, with a general decrease in fluorescence following week one across each salinity treatment. This trend was also observable in HSC; however, each salinity treatment presented an initial peak in metabolic activity in week one, a minimum in week two, then an increase again in week three with the largest of these increases from week two to three observed in 35PSU. Similarly, 35PSU produced the outstanding result within EGC, as there was an increase in respiration in week three which was not observable in the other treatments. While not conclusive, it seems that EGC present a respiration rate slightly lower than HSC; however, there is too much variability within this data to definitively claim this (Fig. 2).

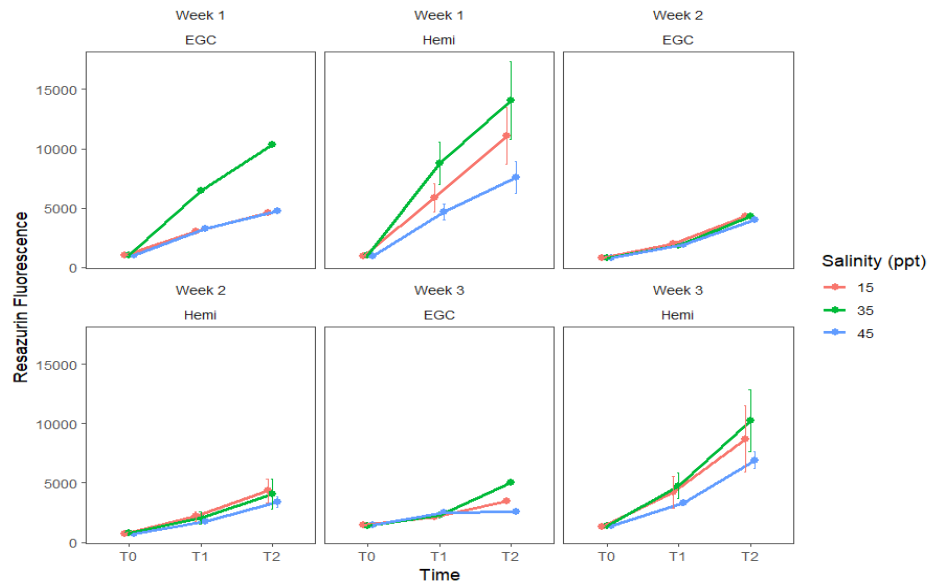


Figure 2. Resorufin fluorescence measured following the use of resazurin to measure respiration in each individual crab across salinity treatments across weeks of experiment. Crabs were divided by species (European green crabs (“EGC”) and hairy shore crabs (“Hemi”). Error bars represent standard error in respective sample groups.

EGCs presented a gradual increase in osmolality as salinity increased, with the 35PSU treatment crab hovering right around the week one HSC baseline. In contrast, the 15PSU HSCs presented an osmolality close to the week one baseline, while 35 and 45PSU treatments were each above the baseline. HSCs in the 45PSU treatment presented a significantly higher osmolality than the baseline, while all other relationships were statistically inconclusive (two-sided t-test, $p\text{-value} = 6.79\text{E-}3$, $df = 3$). Crabs in the 45PSU treatment were farthest away from the baseline value in both species. EGCs seemed to vary less about the baseline compared to HSCs (Fig. 3).

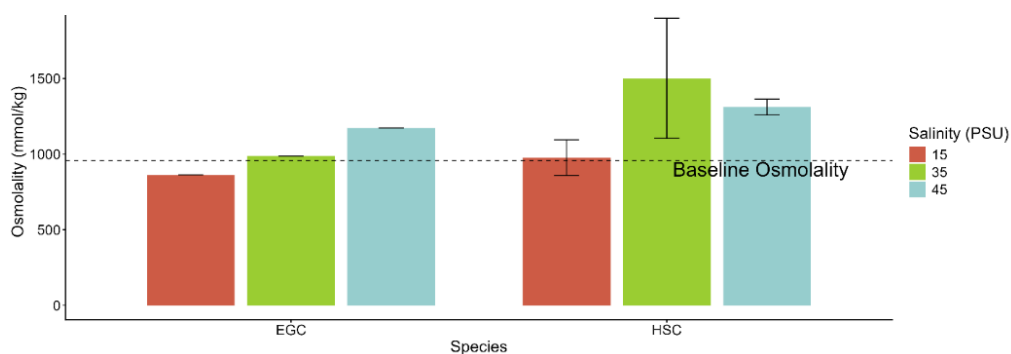


Figure 3. Osmolality (mmol/kg) of hemolymph after 3 weeks of sustained salinity treatments indicated in the legend (15, 35, 45 PSU). One crab per treatment was possible for EGC measurements, and therefore no standard error was calculated. HSC treatments had 3 crabs for 15 and 35 PSU, and 4 crabs for 45 PSU with standard error bars given to those populations. Baseline osmolality measurement (dashed line) taken from 2 averaged, non-experimental EGCs before the start of the experiment.

Discussion

Shorter righting time observed in stressful conditions

Righting time can be used as a tool for understanding how stressful an environment is for a crab. Righting time tends to increase as crabs reach environmental condition limits (Lagerspetz & Vainio, 2006). It was hypothesized that lower salinity treatments such as 15PSU would be more stressful to these crabs than higher salinities, which is reflected in the observed crab's righting times. Crabs in the 15PSU treatment tended to have a lower righting time, indicating that they could be undergoing larger amounts of biological stress. Additionally, the mass-adjusted righting time favoring EGCs may indicate that they would be able to outcompete EGCs for necessary resources in stressful conditions.

Heightened respiration in ambient seawater salinities

The general trend of observing a respiration peak in week one, followed by a minimum in week two, then an increase in week three is likely due to the stress of a new, non-natural habitat, the subsequent adjustment to the new environment, then an increase in stress due to factors such as starvation, respectively. Contrary to the study's hypothesis, crabs in the 35PSU treatment displayed slightly elevated respiration levels. Other studies have found that respiration levels are lowest in ambient seawater, which is the opposite of what was observed in this study. This may be due to the molt of the EGC that occurred in 35PSU prior to week three when elevated respiration spikes were observed. Molting can release a heightened olfactory signal detectible by the EGCs contained within the same tub, causing a stress response and therefore higher respiration (Grasso & Basil, 2002). In contrast, molting is a highly vulnerable process, which may increase stress/respiration in the crab that underwent it (Taylor et al., 2007).

European green crabs show less osmolality variance about the baseline value

As displayed by the data, HSCs cannot osmoregulate as effectively as EGCs. This was an expected result, as highly invasive species must be able to adapt to a wide range of environmental stressors, relative to a native species that is used to a narrow range of conditions. This finding has larger implications for what population dynamics between these two studied species may look like in the futuristic, fresher Puget Sound.

The work completed in this study provides multiple physiological and temporal responses between two competitive species according to an environmental condition that is expected to change in the distant future due to climate change. This study found that the prolonged freshening of Puget Sound will likely cause stress to both species; however, EGCs may be able to regulate their physiology in a more effective way under these conditions, likely due to their invasive nature. However, some findings in this study went against what previous research has found, emphasizing the natural variability in biological responses. This is why it is important to

continue characterizing how predicted environmental changes will impact the health, physiology, and interactions of organisms.

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